§ Poisson structure on S_f via reduction.

- Decompose $\mathfrak{g} = \bigoplus_{i\in\mathbb{Z}} \mathfrak{g}_i$ – h-weight decomposition.
- Consider $\mathfrak{g}_+ = \bigoplus_{i<0} \mathfrak{g}_i$ – nilp. Lie subalgebra.
- Define $\omega_s : \Lambda^2\mathfrak{g}_1 \longrightarrow \mathbb{C}$

$$x\wedge y \longmapsto \chi([x,y]) = \langle f, [x,y]\rangle$$

Prop. ω_s is a symplectic 2-form on $\mathfrak{g}_1$.

Set $\mathfrak{m} = \ell \oplus \bigoplus_{i=2}^{\infty} \mathfrak{g}_i$. Set $M\subset G$: $\mathrm{Lie}(M)=\mathfrak{m}$.

- $\mathfrak{m}$ is a nilpotent Lie algebra.
- Consider restriction map

$$\mu : \mathfrak{g}^* \longrightarrow \mathfrak{m}^*.$$

Set $\chi \in \mathfrak{m}^*$: $\chi : m \longmapsto \langle f, m\rangle$.

Then $\mu^{-1}(\chi) = \chi + \mathfrak{m}^\perp$.

- Note $\chi([\mathfrak{m},\mathfrak{m}]) \subset \langle f, [\ell,\ell] + \bigoplus_{i\geq 2}\mathfrak{g}_i\rangle = 0 \Rightarrow$

$\Rightarrow$ χ is fixed by coadjoint action of $M \Rightarrow \mathcal{O}_M(\chi) = \{\chi\}$.

Theorem [Gan-Ginzburg]

The action map $M\times S_f \xrightarrow{\sim} \chi + \mathfrak{m}^\perp$ is an isomorphism of varieties.

In other words: $\mathfrak{g}^* /\!/_{\mathcal{O}_M(\chi)} M = \mu^{-1}(\mathcal{O}_M(\chi))/M = \chi + \mathfrak{m}^\perp/M \cong S_f$,

Scheme of the proof.

We have $\gamma : SL_2(\mathbb{C}) \longrightarrow G \Rightarrow \rho : \mathbb{C}^\times \curvearrowright \mathfrak{g}$, $t : x \longmapsto t^{-2}\mathrm{Ad}_{\gamma\left(\begin{smallmatrix} t^{-1} & 0 \\ 0 & t\end{smallmatrix}\right)}(x)$.

- ρ contracts S_f to f.
- ρ preserves $\chi + \mathfrak{m}^\perp (= f + \mathfrak{m}^\perp)$.
- Equivariance:

$$M\times S_f \xrightarrow{\quad\alpha\quad} f+\mathfrak{m}^\perp$$

$$\left(\mathrm{Ad}_{\gamma\left(\begin{smallmatrix} t^{-1} & 0 \\ 0 & t\end{smallmatrix}\right)}(g), \rho(t)(x)\right) \longrightarrow \rho(t)(\alpha(g,x)).$$

- Differential at $(1,f)$:

$d\alpha(\mathfrak{m}\oplus \mathcal{Z}_\mathfrak{g}(e)) = [\mathfrak{m},f] \oplus \mathcal{Z}_\mathfrak{g}(e)$. $\dim(\mathfrak{m}^\perp) = \dim(\mathfrak{g}) - \dim(\mathfrak{m}) = \dim(\mathfrak{g}) -$
$-\frac{1}{2}\dim(\mathfrak{g}_1) - \sum_{i\geq 1}\dim(\mathfrak{g}_i) = \dim(\mathfrak{m}) + \dim(\mathfrak{g}_1) + \dim(\mathfrak{g}_0) =$
$= \dim(\mathfrak{m}) + \dim(\mathcal{Z}_\mathfrak{g}(e)) \Rightarrow d\alpha$ – isomorphism.

Note $[\mathfrak{m},f]\cap \mathcal{Z}_\mathfrak{g}(e) = \{0\}$

- Lemma [Gan-Ginzburg] Let X_1, X_2 be affine, smooth, irreducible.
Let $\alpha : X_1 \longrightarrow X_2$ be a $\mathbb{C}^\times$ equivariant map, $\mathbb{C}^\times \curvearrowright X_1$, $\mathbb{C}^\times \curvearrowright X_2$-
-contracting actions. Let $p\in X_1$ be the fixed point.

If $d_p\alpha : T_pX_1 \longrightarrow T_{\alpha(p)}X_2$ is an isomorphism,
Then α is an isomorphism of varieties. ∎

Example:

$$S_f = \left\{ X = \begin{pmatrix} -x_0 & x_3 & x_1 \\ 1 & -x_0 & 0 \\ 0 & x_2 & 2x_0 \end{pmatrix} \middle| x_i \in \mathbb{C} \right\}.$$

$\mathfrak{g}_1 = \langle E_{13}, E_{32}\rangle$

$\omega_s : E_{13}\wedge E_{32} \longrightarrow 1.$

$\ell = \langle E_{13}\rangle$

$\mathfrak{m} = \langle E_{13}, E_{12}\rangle$

$$M = \left\{ \begin{pmatrix} 1 & a & b \\ 0 & 1 & 0 \\ 0 & 0 & 1\end{pmatrix} \middle| a,b\in\mathbb{C}\right\} \cong \mathbb{C}^2.$$

$$\chi + \mathfrak{m}^\perp = \left\{ \begin{pmatrix} \xi_{11} & \xi_{12} & \xi_{13} \\ 1 & \xi_{22} & \xi_{23} \\ 0 & \xi_{32} & -\xi_{11}-\xi_{22}\end{pmatrix}\right\}.$$

$$\begin{pmatrix} \xi_{11} & \xi_{12} & \xi_{13} \\ 1 & \xi_{22} & \xi_{23} \\ 0 & \xi_{32} & -\xi_{11}-\xi_{22}\end{pmatrix} \xmapsto[\begin{pmatrix} 1 & a & b \\ 0 & 1 & 0 \\ 0 & 0 & 1\end{pmatrix}]{} \begin{pmatrix} \xi_{11}+a & \xi_{12}+a(\xi_{22}-\xi_{11})+b\xi_{32} & \xi_{13}+a\xi_{23}+b(-2\xi_{11}-\xi_{22}-a) \\ 1 & \xi_{22}-a & \xi_{23}-b \\ 0 & \xi_{32} & -\xi_{11}-\xi_{22}\end{pmatrix}$$

$a = \frac{\xi_{22}-\xi_{11}}{2}$; $b = \xi_{23} \Rightarrow$

$$\begin{pmatrix} \frac{\xi_{11}+\xi_{22}}{2} & \xi_{12}+\frac{(\xi_{22}-\xi_{11})^2}{2} - \xi_{23}\xi_{32} & \xi_{13}+\xi_{23}(-2\xi_{11}-\xi_{22}) \\ 1 & \frac{\xi_{11}+\xi_{22}}{2} & 0 \\ 0 & \xi_{32} & -\xi_{11}-\xi_{22}\end{pmatrix}$$

$\Rightarrow$ $x_0 = -\frac{\xi_{11}+\xi_{22}}{2}$; $x_1 = \xi_{13} - \xi_{23}(2\xi_{11}+\xi_{22})$,

$x_2 = \xi_{32}$; $x_3 = \xi_{12} + \frac{(\xi_{22}-\xi_{11})^2}{4} + \xi_{23}\xi_{32}$

Using $\{\xi_{ij}, \xi_{k\ell}\} = -(\delta_{jk}\xi_{i\ell} - \delta_{i\ell}\xi_{kj})$ w/ $\sum_{i=1}^{3}\xi_{ii} = 0$ obtain

$\{x_0, x_1\} = \frac{x_1}{2}$; $\{x_0, x_2\} = -\frac{x_2}{2}$; $\{x_0, x_3\} = 0$

$\{x_1, x_2\} = 9x_0^2 - x_3$; $\{x_1, x_3\} = 3x_0x_1$

$\{x_2, x_3\} = -3x_0x_2.$

Casimir functions:

$C_1 = x_3 + 3x_0^2$; $C_2 = 2x_0^3 - 2x_0x_3 + x_1x_2.$

§ Poisson structure via induction.

Recall that we have a Poisson bracket on $\mathfrak{g}^*$.

Claim

- Given a Poisson mfd X for any point $x\in X$ consider the subspace

$F_x = \{ \{f,\cdot\}_x \mid f\in C^\infty(X)\} \subset T_xX.$

- Define $\omega\in\Lambda^2F_x$ by $\omega(\{f,\cdot\}_x, \{g,\cdot\}_x) = \{f,g\}_x$.
- Claim $\forall x$ locally there exists $L_x\subset X$ (locally closed submfd).

  $x\in L_x$, $T_yL_x = F_y$ $\forall y\in L_x$.

  Claim $\omega|_{L_x}$ is symplectic.

- Such maximal L_x is called symplectic leaf.

$$X = \bigsqcup_{L\text{-sympl. leafs}} L.$$

Theorem W/ KKS bracket

- Symplectic leafs of coadjoint action on $\mathfrak{g}^*$ are the coadjoint orbits.
- The corresponding symplectic form can be computed as follows: $\omega_\xi(\mathrm{ad}^*_x(\xi), \mathrm{ad}^*_y(\xi)) = \langle \xi, [x,y]\rangle (= \{x,y\}(\xi)).$

Prop. $\forall x\in S_f$

- $S_f\cap\mathcal{O}_x$ is a smooth locally closed submfd in $\mathfrak{g}$.
- $\tilde\omega = \omega_{\mathcal{O}_x}|_{S_f\cap\mathcal{O}_x}$ is non-degenerate.

Proof. To do. ∎

Def. $\forall x\in S_f$ we have $\overset{\text{non-degenerate}}{\tilde\omega} \in \Lambda^2T^*_x(S_f\cap\mathcal{O}_x)$,

Then define $\tilde\Pi = \iota\circ(\tilde\omega^*)^{-1}$, $\iota_*:\Lambda^2T_x(S_f\cap\mathcal{O}_x) \hookrightarrow \Lambda^2T_xS_f$.

Claim: $\tilde\Pi$ is a Poisson bracket.

DrawNote